

Bootstrap 5 Practicals with Answers

CSS Framework | Responsive Web Design

Grid System & Layout · Navbar & Components · Cards, Modal & Tabs · Forms & Validation · Carousel, Accordion & Toast

Practical 1: Grid System & Layout

Aim: Build responsive layouts using Bootstrap's 12-column grid system.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet"
    href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
</head>
<body class="bg-light p-4">
  <div class="container">
    <h2 class="text-center mb-4">Bootstrap Grid System</h2>

    <!-- Equal columns -->
    <div class="row g-3 mb-3">
      <div class="col"><div class="bg-primary text-white p-3 rounded text-center">col</div></div>
      <div class="col"><div class="bg-primary text-white p-3 rounded text-center">col</div></div>
      <div class="col"><div class="bg-primary text-white p-3 rounded text-center">col</div></div>
    </div>

    <!-- Responsive: 12 -> 6 -> 4 -->
    <div class="row g-3 mb-3">
      <div class="col-12 col-md-6 col-lg-4">
        <div class="bg-success text-white p-3 rounded text-center">col-12 / md-6 / lg-4</div>
      </div>
      <div class="col-12 col-md-6 col-lg-4">
        <div class="bg-success text-white p-3 rounded text-center">col-12 / md-6 / lg-4</div>
      </div>
      <div class="col-12 col-md-12 col-lg-4">
        <div class="bg-success text-white p-3 rounded text-center">col-12 / md-12 / lg-4</div>
      </div>
    </div>

    <!-- Offset -->
    <div class="row g-3">
      <div class="col-md-4 offset-md-4">
        <div class="bg-warning text-dark p-3 rounded text-center">
          col-4 offset-4 (centered)
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

Explanation:

Bootstrap grid: **container** (centered wrapper), **row** (flex row), **col-{n}** (column spanning n of 12). Breakpoints: xs (default), **sm** (576px), **md** (768px), **lg** (992px), **xl** (1200px). **g-{n}**: gutter/gap. **offset-md-{n}**: push column right by n columns.

Output: Three equal columns, then responsive 3-col layout that stacks on mobile, and a centered offset column.

Practical 2: Navbar & Components

Aim: Build a responsive navbar and use Bootstrap UI components.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet"
    href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
</head>
<body>
  <!-- Responsive Navbar -->
  <nav class="navbar navbar-expand-lg navbar-dark bg-dark px-3">
    <a class="navbar-brand fw-bold text-warning" href="#">PlacePortal</a>
    <button class="navbar-toggler" type="button">
```

```

        data-bs-toggle="collapse" data-bs-target="#navMenu">
    <span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="navMenu">
    <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#">Jobs</a></li>
        <li class="nav-item">
            <a class="btn btn-primary ms-2" href="#">Login</a>
        </li>
    </ul>
</div>
</nav>

<div class="container mt-4">
    <!-- Badges, Alerts, Buttons -->
    <p>Status:
        <span class="badge bg-success">Active</span>
        <span class="badge bg-warning text-dark">Pending</span>
        <span class="badge bg-danger">Closed</span>
    </p>
    <div class="alert alert-info alert-dismissible fade show" role="alert">
        <strong>Info!</strong> Applications close on Friday.
        <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
    </div>
    <div class="d-flex gap-2">
        <button class="btn btn-primary">Primary</button>
        <button class="btn btn-outline-secondary">Outline</button>
        <button class="btn btn-success btn-sm">Small</button>
        <button class="btn btn-danger btn-lg">Large</button>
    </div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Explanation:

navbar-expand-lg: collapses to hamburger below lg. **data-bs-toggle='collapse'**: JS-powered toggle. **ms-auto**: margin-start auto (push to right). **badge**: colored label. **alert alert-{type}**: notification box. **btn btn-{variant}**: styled buttons. **btn-outline-{color}**: bordered button.

Output: Dark navbar collapses on mobile. Info alert has close button. Four button variants shown side by side.

Practical 3: Cards, Modal & Tabs

Aim: Use Bootstrap card component, modal dialog, and tab navigation.

Code:

```

<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet"
        href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
</head>
<body class="p-4 bg-light">
    <!-- Tabs -->
    <ul class="nav nav-tabs mb-3" id="myTab">
        <li class="nav-item">
            <button class="nav-link active" data-bs-toggle="tab"
                data-bs-target="#home">Home</button>
        </li>
        <li class="nav-item">
            <button class="nav-link" data-bs-toggle="tab"
                data-bs-target="#jobs">Jobs</button>
        </li>
    </ul>
    <div class="tab-content">
        <div class="tab-pane fade show active" id="home">

```

```

<!-- Card -->
<div class="card" style="max-width:320px">
  <div class="card-header bg-primary text-white">Job Opening</div>
  <div class="card-body">
    <h5 class="card-title">Python Developer</h5>
    <p class="card-text text-muted">3-6 months internship. Remote.</p>
    <button class="btn btn-primary" data-bs-toggle="modal"
      data-bs-target="#applyModal">Apply Now</button>
  </div>
  <div class="card-footer text-muted">Posted: Today</div>
</div>
</div>
<div class="tab-pane fade" id="jobs">
  <p class="text-muted">More jobs coming soon...</p>
</div>
</div>

<!-- Modal -->
<div class="modal fade" id="applyModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Apply for Python Developer</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">
        <input type="text" class="form-control mb-2" placeholder="Your Name">
        <input type="email" class="form-control" placeholder="Email">
      </div>
      <div class="modal-footer">
        <button class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
        <button class="btn btn-primary">Submit</button>
      </div>
    </div>
  </div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Explanation:

nav nav-tabs: tab navigation. **tab-pane fade show active**: tab content panel. **card / card-header / card-body / card-footer**: card structure. **modal fade**: hidden dialog. **data-bs-toggle='modal' data-bs-target='#id'**: trigger modal. All interactivity via HTML data attributes, no custom JS needed.

Output: Two tabs switch content. Card in first tab has header, body, footer. Apply button opens a modal dialog with a form.

Practical 4: Forms & Validation

Aim: Build styled Bootstrap forms with validation states.

Code:

```

<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet"
    href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
</head>
<body class="p-4">
  <div class="card shadow-sm mx-auto" style="max-width:500px">
    <div class="card-body">
      <h4 class="card-title mb-4">Student Registration</h4>
      <form class="needs-validation" novalidate
        onsubmit="this.classList.add('was-validated');return false;">

        <div class="mb-3">
          <label class="form-label">Full Name</label>
          <input type="text" class="form-control" required

```

```

        placeholder="Enter full name">
        <div class="invalid-feedback">Name is required.</div>
        <div class="valid-feedback">Looks good!</div>
    </div>

    <div class="mb-3">
        <label class="form-label">Email</label>
        <input type="email" class="form-control" required>
        <div class="invalid-feedback">Valid email required.</div>
    </div>

    <div class="mb-3">
        <label class="form-label">Branch</label>
        <select class="form-select" required>
            <option value="">Choose...</option>
            <option>Computer Engineering</option>
            <option>IT</option>
            <option>Mechanical</option>
        </select>
    </div>

    <div class="mb-3 form-check">
        <input type="checkbox" class="form-check-input" required>
        <label class="form-check-label">I agree to Terms</label>
        <div class="invalid-feedback">Must agree to continue.</div>
    </div>

    <button type="submit" class="btn btn-primary w-100">Register</button>
</form>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Explanation:

form-control: styled input. **form-label**: label styling. **form-select**: styled dropdown. **form-check** / **form-check-input**: checkbox/radio. **needs-validation** + **novalidate**: custom Bootstrap validation. **was-validated** class (added on submit) triggers **invalid-feedback** / **valid-feedback** messages. **required** attribute drives validation.

Output: Form with name, email, select, checkbox. On submit, red invalid messages appear for empty fields and green for valid ones.

Practical 5: Carousel, Accordion & Toast

Aim: Use Bootstrap interactive components: carousel, accordion, and toast.

Code:

```

<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet"
        href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
</head>
<body class="p-4 bg-light">
    <!-- Accordion -->
    <div class="accordion mb-4" id="faq">
        <div class="accordion-item">
            <h2 class="accordion-header">
                <button class="accordion-button" data-bs-toggle="collapse"
                    data-bs-target="#q1">What is Bootstrap?</button>
            </h2>
            <div id="q1" class="accordion-collapse collapse show"
                data-bs-parent="#faq">
                <div class="accordion-body">Bootstrap is a free CSS framework for
                    building responsive websites quickly.</div>
            </div>
        </div>
    </div>

```

```

<div class="accordion-item">
  <h2 class="accordion-header">
    <button class="accordion-button collapsed" data-bs-toggle="collapse"
      data-bs-target="#q2">How to use it?</button>
  </h2>
  <div id="q2" class="accordion-collapse collapse"
    data-bs-parent="#faq">
    <div class="accordion-body">Add Bootstrap CDN link to your HTML head
      and use predefined classes.</div>
  </div>
</div>
</div>

<!-- Toast trigger -->
<button class="btn btn-success"
  onclick="new bootstrap.Toast(document.getElementById('myToast')).show()">
  Show Toast
</button>

<!-- Toast -->
<div class="toast-container position-fixed bottom-0 end-0 p-3">
  <div id="myToast" class="toast align-items-center text-bg-success">
    <div class="d-flex">
      <div class="toast-body">Form submitted successfully!</div>
      <button class="btn-close btn-close-white ms-auto me-2 m-auto"
        data-bs-dismiss="toast"></button>
    </div>
  </div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Explanation:

accordion / accordion-item / accordion-button: collapsible FAQ panels. **data-bs-parent:** ensures only one panel open at a time. **toast:** small notification that auto-dismisses. **toast-container position-fixed bottom-0 end-0:** positions toast at bottom-right. **new bootstrap.Toast(el).show():** programmatically show toast via JS.

Output: Accordion FAQ with first item expanded. Clicking other collapses previous. Toast notification pops at bottom-right when button clicked.

-- End of Practicals --